Computer Networks – Important Topics

Beginner-friendly study notes for technical interviews, aptitude tests, and exams.

1. Computer Network Basics

- Definition and purpose of computer networks
- Types of networks: PAN, LAN, MAN, WAN
- Network topologies: Bus, Star, Ring, Mesh, Tree, Hybrid
- Network devices: Hub, Switch, Router, Bridge, Repeater, Gateway, Modem, Access Point
- Client-server and peer-to-peer networks
- Bandwidth, throughput, latency, jitter, and transmission delay

2. Network Models

- OSI Reference Model – 7 layers
- Functions of Physical, Data Link, Network, Transport, Session, Presentation, and Application layers
- TCP/IP model and its layers
- OSI vs TCP/IP
- Encapsulation and decapsulation
- Protocol Data Units (bits, frames, packets, segments)

3. Physical Layer

- Transmission media: guided and unguided
- Twisted pair, coaxial cable, fiber optic cable
- Radio, microwave, infrared, satellite communication
- Analog vs digital signals
- Bit rate and baud rate
- Multiplexing: FDM, TDM, WDM

4. Data Link Layer

- Framing and frame structure
- MAC addressing
- Error detection: parity, checksum, CRC
- Error correction basics
- Flow control
- Stop-and-Wait and Sliding Window protocols
- ARQ: Stop-and-Wait ARQ, Go-Back-N, Selective Repeat
- Ethernet and CSMA/CD
- MAC address vs IP address

5. Network Layer

- IPv4 addressing
- Public vs private IP addresses
- IPv4 address classes (basic concept)
- Subnetting and subnet masks
- CIDR notation
- Default gateway
- IPv6 basics and IPv4 vs IPv6
- ARP, ICMP, DHCP
- Routing and forwarding
- Static vs dynamic routing
- Distance Vector and Link State routing
- Routing protocols: RIP, OSPF, BGP

6. Transport Layer

- Purpose of the Transport Layer
- TCP vs UDP
- TCP three-way handshake
- TCP connection termination
- TCP reliability, acknowledgements, retransmission
- Flow control and congestion control
- Port numbers and sockets
- Well-known ports: HTTP 80, HTTPS 443, FTP 21, SSH 22, DNS 53

7. Application Layer

- HTTP and HTTPS
- DNS and domain name resolution
- FTP and SFTP
- SMTP, POP3, IMAP
- SSH and Telnet
- DHCP
- Web browsers and web servers
- URL, domain name, IP address, and DNS relationship

8. Network Security Basics

- Authentication and authorization
- Encryption and decryption
- Symmetric vs asymmetric encryption

- Digital signatures and certificates
- Firewall
- VPN
- HTTPS and TLS
- Common attacks: phishing, DoS/DDoS, spoofing, sniffing, man-in-the-middle

9. Important Interview Comparisons

- Hub vs Switch vs Router
- Switch vs Router
- MAC address vs IP address
- IPv4 vs IPv6
- TCP vs UDP
- HTTP vs HTTPS
- FTP vs SFTP
- OSI vs TCP/IP
- Connection-oriented vs connectionless communication
- Unicast vs multicast vs broadcast
- Bandwidth vs throughput
- Flow control vs congestion control
- Static routing vs dynamic routing

10. High-Priority Topics for Exams/Interviews

- OSI 7 layers and their functions
- TCP/IP model
- TCP three-way handshake
- TCP vs UDP
- IP addressing and subnetting
- MAC address and ARP
- DNS working
- DHCP working
- HTTP vs HTTPS
- Network devices
- Routing basics
- Framing and error detection
- Flow and congestion control
- Common port numbers
- Basic network security

Quick OSI Layer Order

Layer	Name	Main idea
7	Application	Network services to applications
6	Presentation	Translation, encryption, compression
5	Session	Session establishment and management
4	Transport	End-to-end delivery, TCP/UDP
3	Network	IP addressing and routing
2	Data Link	Frames and MAC addressing
1	Physical	Bits and transmission media